

TAMOJIT BERA

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EDUCATION

BACHELOR OF SCIENCE, COMPUTER SCIENCE

September 2022 - May 2026

University of Minnesota - Twin Cities — GPA: 3.947

MN, USA

Dean's List - Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025

Global Excellence Scholarship

Coursework: Algorithms, Operating Systems, Database Systems, Program Design, AI, ML, Machine Architecture, Computer Networks, Linear Algebra, Full-Stack Web Development, Cryptography, Computer Security, Natural Language Processing

EXPERIENCE

UNDERGRADUATE RESEARCH ASSISTANT

January 2025 - Present

Minnesota Robotics Institute

MN, USA

Supervisor: Prof. Maria Gini

- Engineered AI-driven real-time object detection on NVIDIA Jetson to autonomously detect, track, and deter wildlife
- Awarded "Excellence in Small Farms Tech" (Air Category) at the Farm Robotics Challenge, 2025

UNDERGRADUATE TEACHING ASSISTANT

September 2023 - Present

CSCI 1133, CSCI 2041, CSCI 2033 - University of Minnesota

MN, USA

- Supervised groups of 200+ students, offering assistance in Programming Principles and Linear Algebra

RESEARCH INTERN

June 2025 - August 2025

Cyber Physical Systems Lab, Georgia Institute of Technology

GA, USA

Supervisor: Dr. Animesh Chhotaray

- Designed a vision-based anomaly detection pipeline for DED printing

SOFTWARE ENGINEERING INTERN

June 2025 - August 2025

Attention.ad

DE, USA

- Spearheaded GenAI driven recommendation systems for blockchain-based creator platforms

SOFTWARE ENGINEERING INTERN

June 2024 - August 2024

ATHRV Cloud

OD, INDIA

- Designed GenAI frameworks for optimizing and classifying bug reports in embedded C/C++ codebase

APP DEVELOPER

November 2023 - April 2024

Google Developer Student Club, University of Minnesota

MN, USA

- Developed an Android app gamifying eco-friendly choices using Java and Kotlin on Android Studio

PROJECTS

AES-CBC PADDING ORACLE ATTACK — Python, Cryptanalysis, AES, PKCS#7

- Implemented Vaudenay's padding oracle attack to decrypt AES-CBC ciphertexts via side-channel analysis
- Analyzed time complexity, attack feasibility, and mitigation via authenticated encryption (AES-GCM)

VISUAL TRANSIT SIMULATOR — Java, Node.js, Gradle, React

- Architected a comprehensive transit system simulating realistic bus/train schedules, routing, and traffic

AI AGENTS FOR CONNECT-4 — Python, NumPy, Matplotlib, Heuristic Analysis

- Developed competitive AI agents using Minimax with Alpha-Beta Pruning and Monte Carlo Tree Search
- Benchmarked over 600 self-play matches to assess strategic depth and computational efficiency

IMAGE MATCHING SYSTEM — C, POSIX Threads, Socket, Mutex

- Implemented a multithreaded client-server model using sockets and mutexes for concurrent requests
- Developed efficient matching algorithms and modular database storage for scalable performance

PROFESSIONAL COURSES

AWS: IT and AWS Cloud, Cloud Technical Essentials, Developing Applications on AWS

IBM: Code Generation and Optimization Using IBM Granite

SKILLS

Languages: Python, Java, Kotlin, C, C++, OCaml, SQL, JavaScript, x86-64 Assembly

Frameworks & Libraries: React, Node.js, Express.js, OpenCV, DeepSORT, Scikit-learn, Pandas, NumPy, Matplotlib

Developer Tools & Platforms: Android Studio, Docker, AWS, GitHub, Grafana, InfluxDB, IBM Granite, Firebase

Technologies & Systems: Computer Vision, NVIDIA Jetson, Static Code Analysis, Embedded Systems, Telemetry Systems